Frequently Bought Together

...and the three things it can't do

60 minutes · illustrative portfolio scenario · simulated telemetry · nothing deployed

AI Engineering · business engagement

Would you have shipped this?

A recommendation from the live table, computed from the retail dataset

WHITE METAL LANTERN → WHITE HANGING HEART T-LIGHT HOLDER

63 baskets, confidence 0.19, lift 1.65 — looks like a finding

...until you notice **the same t-light holder is recommended for 79 products**

It is in a large share of every basket, so it pairs with everything.

Hold that thought

Every number here is from the live table.

By 0:30 you'll be able to say in one sentence why this row is weak.

Popularity is not affinity. Always ask what else that product is recommended for.

Reading a recommendation

One real row, one number at a time

CHILDS GARDEN SPADE PINK → CHILDS GARDEN SPADE BLUE

40

baskets

Your sample size. Read this first — a huge lift on 8 baskets is noise.

0.48

confidence

Of baskets with the pink spade, 48% also had the blue one.

95

lift

95× the independent-purchase expectation. This is the ranking column.

Back to the first row: lift 1.65 against 95 here. Same catalogue, different strengths of association.

Two signals — and they are not equal

Check the method column before you build a plan on a row

✓ co_purchase

Built from what people actually bought together.

The real signal — trust it in proportion to the basket count.

Covers about 30% of the catalogue.

! content_tfidf

Built from the product description text.

Nobody has been observed buying these together — they simply read alike.

A starting point, never evidence.

Why keep the weaker signal? Without it 7 products in 10 show an empty panel — and people generalise from the first empty screen they see.

Three things it cannot do

Said first and unprompted — the credibility of everything else depends on it

1. **It cannot tell you why**

   Bunting and paper plates co-occur because both are party purchases — not because one drives the other. Placing them together is sensible. Discounting one to lift the other is not.

2. **It cannot see what you didn't stock**

   A product out of stock for six weeks builds its pairings from the weeks it was available. It will look less connected than it is.

3. **It cannot decide anything**

   Nothing writes to a trading system. Every recommendation is a suggestion a person accepts or rejects. That is a design decision, not a limitation.

Reading the simulated adoption report

Simulated weekly reach — % of 62 fictional users active that week

Illustrative workshop dates

These changes are generated examples.
They do not measure workshop impact.

With real telemetry, check sustained use, missing data and a comparison group before attributing change to training.

Exercise: identify the evidence needed to distinguish sustained use from a short-lived spike.